Precision Pharmacology: Project Documentation & Notes

1. Project Overview & Business Value

The cost of bringing a single new drug to market is currently estimated at over $2 billion, taking up to 10-15 years. However, "Drug Repurposing", finding new uses for existing or previously tested compounds, can drastically reduce this cost and timeline.

The goal of this project was to ingest millions of genetic data points, architect a relational SQL database, and identify "Precision Leads": chemical compounds that surgically alter disease genes without causing toxic off-target effects.

2. Biological Context (Understanding the Data)

To understand the code, one must understand the biology it represents. This project utilizes the **LINCS L1000 Dataset** from the Broad Institute.

> **The Landmark Genes:** Humans have roughly 20,000 genes. Measuring all of them is too expensive. The L1000 dataset only measures 978 "Landmark" genes. Because these genes are highly connected, if we know what these 978 are doing, we can infer the behavior of the rest of the cell.
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> **MCF7 Cell Line:** The experiments we filtered for were performed on MCF7 cells, which are a standard model for human breast cancer.
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> **Z-Scores (The Metric of Impact):** The values in our database are Z-scores. A Z-score of 0 means the drug did nothing. A Z-score of +3.0 means the drug strongly "turned on" (up-regulated) a gene. A Z-score of -3.0 means it strongly "turned off" (down-regulated) a gene. In biology, an absolute Z-score above 2.0 is considered statistically significant.

3. Data Engineering & Architecture

Overcoming the "Wide Data" Problem

The raw data was provided in a .gctx format, resembling a massive spreadsheet with 978 rows (genes) and 14,861 columns (drug experiments). SQL databases perform terribly with 14,000 columns.

> **The Solution (Melt/Stack):** Using Python's Pandas library, the data was "melted" into a "Long" format, generating over 14.5 million individual observations.

Dimensionality Reduction: We filtered out all Z-scores between -2.0 and 2.0, removing biological "noise" and reducing the dataset size to roughly 1 million significant interactions, ensuring high-speed SQL queries.

The Relational SQL Schema

To emulate an enterprise data warehouse, the flat data was split into a **Relational Schema** using Primary and Foreign Keys:

compound_metadata (Table 1): Contains drug names, Mechanisms of Action (MoA), and structural chemistry strings (SMILES).

genetic_signatures (Table 2): Contains the actual experimental Z-scores.

All queries to answer the business questions utilized SQL JOIN statements to link the biological identity of the drug to its genetic impact.

4. Analytical Methodology (The 10 Business Questions)

We utilized SQL to translate strategic pharmaceutical objectives into data queries:

Signature Reversal: Finding drugs that "flip" a known disease state (e.g., turning on genes that a disease turns off).

MoA Discovery: Matching the genetic fingerprints of unknown lab codes to well-understood, FDA-approved drugs.

Safety Screening: Counting extreme Z-scores ($|Z| > 4.0$) to flag drugs with high transcriptomic volatility (side effects).

Master Regulators: Identifying genes (like ATP6V1D) that react most frequently to chemical perturbations.

Precision vs. Power: Finding the "Goldilocks Zone" by charting the maximum potency of a drug against its specificity (how few genes it affects).

5. Web Scraping & Intellectual Property (IP) Profiling

A critical question in pharmacology is Commercial Viability. We successfully identified 10 highly potent drug leads using SQL, but we needed to determine their "Research Maturity." Are these drugs heavily studied (safe for repurposing), or are they undiscovered "Lab Codes" (untapped, highly lucrative Intellectual Property)?

The BeautifulSoup Web-Scraper: We engineered a custom web scraper using Python's requests and BeautifulSoup libraries.

The Literature Footprint Audit: The script systematically looped through our Top 10 Final Leads and physically navigated to the US National Library of Medicine (PubMed). It parsed the static HTML of the search results to extract the exact number of published, peer-reviewed scientific papers for each compound. This allowed us to mathematically categorize our leads into "Established Therapies" versus "Novel Proprietary IP" (0 published papers).